RISHI KIRAN KARNATAKAM
8328388715 — karnatakamrishi@gmail.com
linkedin.com/in/rishi-kiran-karnatakam — github.com/Rishikarnatakam

Summary

Motivated Computer Science student seeking job and internship opportunities to apply my skills in software development and problem-solving. Quick to learn new technologies and adapt to challenges, focusing on contributing to innovative and impactful projects.

Education

NNRESGIDec 2021 – PresentBachelor of Technology in Computer Science and Engineering (GPA: 8.4)

KMIITSep 2019 – Jun 2021Intermediate (10+2 equivalent) (GPA: 9.4)

SAGHSMay 2019Secondary School Certificate (GPA: 9.8)

Skills

- **Programming Languages:** C, Python, Dart, C#
- **Databases:** MySQL, MongoDB
- **Cloud Platforms:** Amazon Web Services (AWS)
- **Machine Learning/AI:** TensorFlow, TensorFlow Lite
- **Version Control:** Git, GitHub
- **Frameworks/Libraries:** Flask, Tkinter, Chess Library, Unity Engine, Flutter
- **DevOps Tools:** Jenkins

Projects

AI-Integrated Plant Disease Detection Mobile AppApr 2025

Built a mobile app for real-time plant disease detection using camera or gallery images.

Integrated a TensorFlow Lite model (Inception V3) for accurate disease prediction with offline support.

Enabled local data storage and automatic syncing to MongoDB Atlas when connected online.

Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.

Technologies: Flutter, Dart, TensorFlow Lite, MongoDB Atlas

AI-Powered Chess Engine with Genetic Algorithm OptimizationAug 2024

Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.

Enhanced strategic depth and move accuracy by 20%.

Achieved a 30% improvement in move generation speed through alpha-beta pruning.

Technologies: Python, Tkinter, Chess Library

Cotton Plant Disease Detection and Classification Using Transfer LearningFeb 2024

Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.

Improved disease detection accuracy by 25% using transfer learning with pre-trained CNN models.

Achieved 90%+ accuracy on test datasets, optimizing performance for agricultural disease diagnosis.

Technologies: Python, TensorFlow

Interactive Solar System Model and 3D Game DevelopmentOct 2023

Developed a 3D interactive solar system model to enhance gamified learning.

Created a first-person 3D game inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.

Technologies: C#, Unity Engine

Awards

- **National Hackathon on Generative AI** – Achieved 2nd place in a National Level Hackathon focused on Generative AI. (Sep 2024)
- **Internal Hackathon on Generative AI** – Achieved 3rd place in a college-hosted Hackathon focused on Generative AI. (Aug 2024)
- **National Hackathon on AR/VR Development** – Ranked in the top 10 at a national-level hackathon focused on AR/VR. (Oct 2023)

Certifications

AWS Academy Graduate - AWS Academy Cloud ArchitectingSep 2024AWS Academy

AICTE Virtual Internship2024Nalla Narasimha Reddy Education Society's Group of Institutions

Python For Data ScienceJan 2025IIT Madras (NPTEL)

The Joy of Computing Using PythonJul 2023IIT Madras (NPTEL)

Supervised Machine Learning: Regression and ClassificationJun 2023Stanford University (Coursera)